

$$(x+y-7)^2 + (\cancel{2xy}y + 3x - 13)^2 = 0$$

→ ~~10x~~

$$x^T V x + 2u^T x + f = 0$$

$$10x^2 + 8xy + 2y^2 - 92x - 40y + 228 = 0$$

$$x^T \begin{pmatrix} a & b \\ b & d \end{pmatrix} x = \cancel{10}x^2 + 8xy + 2y^2$$

$$\Rightarrow a = 10$$

$$b = 4$$

$$c = 2$$

$$\Rightarrow V = \begin{bmatrix} 10 & 4 \\ 4 & 2 \end{bmatrix}$$

$$2u^T x = -92x - 40y$$

$$\Rightarrow u = \begin{bmatrix} -46 \\ -20 \end{bmatrix}$$

$$\Rightarrow f = 228$$

$$\left| \begin{array}{ccc|c} 10 & 4 & -46 & 0 \\ 4 & 2 & -20 & 0 \\ -46 & -20 & 228 & 0 \end{array} \right| = 0 \quad \left\{ \begin{pmatrix} V & u \\ u^T & f \end{pmatrix} x = 0 \right\}$$

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$$C_1 \Rightarrow C_1 - 2C_2$$

$$\left| \begin{array}{ccc|c} 2 & 4 & -46 & 0 \\ 0 & 2 & -20 & 0 \\ -6 & -20 & 228 & 0 \end{array} \right|$$

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$$C_2 \rightarrow C_2 - 2C_1$$

$$\left| \begin{array}{ccc|c} 2 & 0 & -46 & 0 \\ 0 & 2 & -20 & 0 \\ -6 & -8 & 228 & 0 \end{array} \right|$$

$$\left| \begin{array}{ccc} 1 & 0 & -23 \\ 0 & 1 & -20 \\ -3 & -4 & 109 \end{array} \right|$$

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$$\left| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & -20 \\ -3 & -4 & 40 \end{array} \right|$$

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$$\left| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 0 & -20 \\ -3 & 0 & 40 \end{array} \right| = 0$$

$$\begin{aligned}x + y - 7 &= 0 \\ 3x + y - 13 &= 0\end{aligned}$$

$$\begin{aligned}\Rightarrow x + y &= 7 \\ 3x + y &= 13\end{aligned}$$

Augmented Matrix is

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 3 & 1 & 13 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 3R_1$$

$$\left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & -2 & -8 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 1 & 7 \\ 0 & 1 & 4 \end{array} \right]$$

$$R_1 \rightarrow R_1 - R_2$$

$$\left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 4 \end{array} \right]$$

$$\Rightarrow x = 3, y = 4$$